

Regularized Multivariate Two-way Functional Principal Component Analysis

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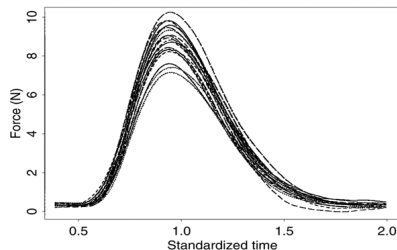
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

Functional Data Analysis:

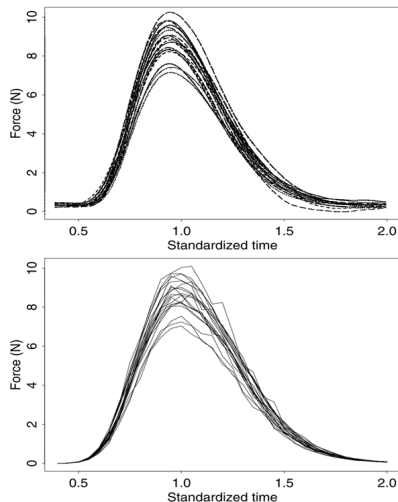
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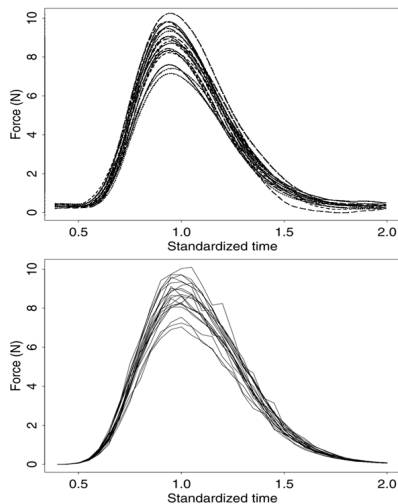
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- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



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- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work

References

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